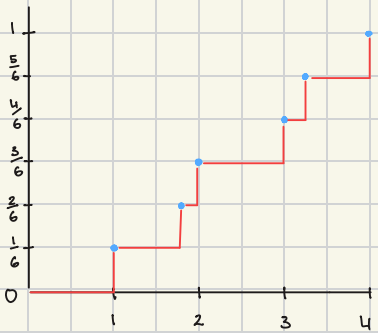
Problem # 1

[1, 1.75, 2, 3, 3.25, 4] $n = 6$

a)



b) $h = 0.2$

1 => [0.8, 1.2]
1.75 => [1.55, 1.95]
2 => [1.8, 2.2]
3 => [2.8, 3.2]
3.25 => [3.05, 3.45]
4 => [3.8, 4.2]

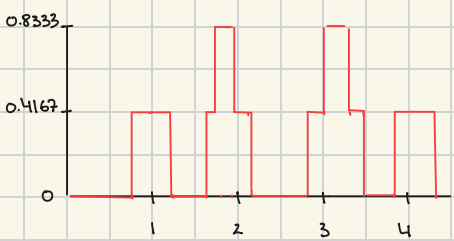
$\frac{1}{2nh} = \frac{1}{2(6)(0.2)}$

$= \frac{1}{2.4}$

$= 0.4167$ = height

overlaps = [1.80, 1.95], [3.05, 3.20]

$\frac{2}{2.5} = 0.8333$



c) $h = 1.84\,(SD)(n^{-0.2})$

$\bar{x} = \frac{1 + 1.75 + 2 + 3 + 3.25 + 4}{6}$

$= \frac{15}{6} = 2.5$

$SD = \sqrt{\frac{\Sigma (x_i - \bar{x})^2}{n-1}}$

$$SD = \frac{(1-2.5)^2 + (1.75-2.5)^2 + (2-2.5)^2 + (3-2.5)^2 + (3.25-2.5)^2 + (4,-2.5)^2}{6-1}$$

$$= \frac{6.125}{5}$$

$$= 1.225$$

$$= \sqrt{1.225}$$

$$= 1.1068$$

$$h = 1.84(1.1068)(6^{-0.2})$$

$$= 1.4232$$

d) $h = 1.423$

1 => [-0.423, 2.423]
1.75 => [0.327, 3.173]
2 => [0.577, 3.423]
3 => [1.577, 4.423]
3.25 => [1.827, 4.673]
4 => [2.577, 5.423]

overlaps = [0.327, 0.577]
[0.577, 1.577]
[1.577, 1.827]
[1.827, 2.423]
[2.423, 2.577]
[2.577, 3.173]
[3.173, 3.423]
[3.423, 4.423]
[4.423, 4.673]

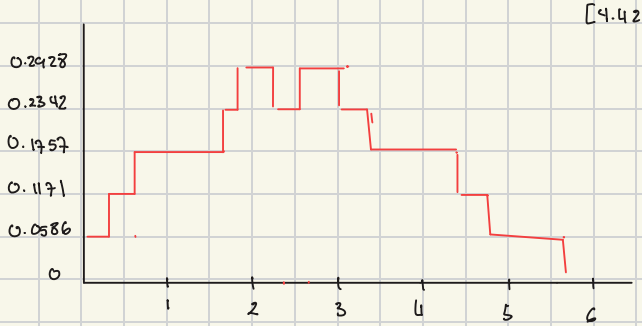
$\frac{1}{216(1.423)} = \frac{1}{17.076} = 0.0586$

$\frac{2}{17.076} = 0.1171$

$\frac{3}{17.076} = 0.1757$

$\frac{4}{17.076} = 0.2342$

$\frac{5}{17.076} = 0.2928$



e) main difference is the bandwidth, larger bandwidth create more overlap, smaller bandwidth less overlap.

Problem #2

$X = [-1, -0.5, 0, 0.5, 1]$, $n = 5$, $h = 0.6$

a) $(x-h, x+h) \Rightarrow [-0.6, 0.6]$

$[-1, \boxed{-0.5, 0, 0.5}, 1]$

$$\hat{f}_h(0) = \frac{3}{2(5)(0.6)} = \frac{3}{6} = 0.5$$

b) $E(\hat{f}_h(x)) = \frac{P(-0.6 < x < 0.6)}{2h} \Rightarrow$ $P(x \leq 0.6) = 0.7257$, $P(x \leq -0.6) = 0.2743$

$0.7257 - 0.2743 = 0.4514 = P(-0.6 < x < 0.6)$

$$\frac{0.4514}{2(0.6)} = 0.3762$$

c) $f(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$

$= \frac{1}{\sqrt{2\pi}} e^{0}$

$= \frac{1}{\sqrt{2}\pi}$

$= 0.3989$

d) $0.3762 \neq 0.3989$

Then KDE is biased at $x = 0$ for $h = 0.6$

and the bias is $(0.3762 - 0.3989) = \boxed{-0.0227}$

e) $h = 0.1$

$P(x \leq 0.1) = 0.4602$

$P(x \leq -0.1) = 0.5398$

$0.5398 - 0.4602 = 0.0796$

$\frac{0.0796}{0.2} = 0.3980$ $\Rightarrow$ as h shrinks, the gap get smaller, making h smaller reduced the bias

f) yes, $\hat{f}_h$ is consistent because as we get more data and h gets smaller, the KDE gets closer to the true density

g) small h, less bias but more noise
large h, less noise, but more bias
⇒ we need consider the tradeoffs

Problem #3

c) $x = [5, 7, 7, 8, 10]$ $n = 5$ $h = 1.5$ $\hat{f}_h(x) = \frac{\sum_{i=1}^{n} I\{|x_i - x| < h\}}{2nh}$ ← count

$|x_i - x| < h$

x_i
$x_1 = 5$
$x_2 = 7$
$x_3 = 7$
...

$\hat{f}_h(6.0) = \frac{3}{2(5)(1.5)} = 0.200$

$\hat{f}_h(6.5) = \frac{4}{15} = 0.267$

$\hat{f}_h(7.0) = \frac{3}{15} = 0.200$

$\hat{f}_h(8.0) = \frac{3}{15} = 0.200$

$\hat{f}_h(8.5) = \frac{1}{15} = 0.067$

b) $z = \frac{x - x_i}{h}$, $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$, $\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} \phi(z)$, $n = 5$, $h = 1.5$

$$\hat{f}_h(6) = \frac{1}{(5)(1.5)} \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{6-5}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{6-7}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{6-7}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{6-8}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{6-5}{1.5}\right)}{2}} \right)$$

$$= \frac{0.3194 + 0.3194 + 0.3194 + 0.1640 + 0.0114}{7} = 0.1512$$

$$\hat{f}_h(6.5) = \frac{1}{(5)(1.5)} \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{6.5-5}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{6.5-7}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{6.5-7}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{6.5-8}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{6.5-5}{1.5}\right)}{2}} \right)$$

$$= \frac{0.2420 + 0.3774 + 0.3774 + 0.2420 + 0.0262}{7.5} = 0.1687$$

$$\hat{f}_h(7) = \frac{1}{(5)(1.5)} \left(\frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{7-5}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{7-7}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{7-7}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{7-8}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}} e^{-\frac{\left(\frac{7-5}{1.5}\right)}{2}} \right)$$

$$= \frac{0.1640 + 0.3989 + 0.3989 + 0.3194 + 0.0540}{7.5} = 0.1780$$

$$\hat{f}_h(8) = \frac{1}{(5)(1.5)}\left(\frac{1}{\sqrt{2\pi}}e^{-\frac{\left(\frac{8-5}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}}e^{-\frac{\left(\frac{8-7}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}}e^{-\frac{\left(\frac{8-7}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}}e^{-\frac{\left(\frac{8-8}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}}e^{-\frac{\left(\frac{8-5}{1.5}\right)}{2}}\right)$$

$$= \frac{0.0540 + 0.3194 + 0.3194 + 0.3989 + 0.1640}{7.5} = 0.1674$$

$$\hat{f}_h(10) = \frac{1}{(5)(1.5)}\left(\frac{1}{\sqrt{2\pi}}e^{-\frac{\left(\frac{10-5}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}}e^{-\frac{\left(\frac{10-7}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}}e^{-\frac{\left(\frac{10-7}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}}e^{-\frac{\left(\frac{10-8}{1.5}\right)}{2}} + \frac{1}{\sqrt{2\pi}}e^{-\frac{\left(\frac{10-5}{1.5}\right)}{2}}\right)$$

$$= \frac{0.0262 + 0.2420 + 0.2490 + 0.3774 + 0.2420}{7.5} = 0.1506$$

c) biggest jump is from 8 to 8.5, $\hat{f}_h(8) = 0.200$ and $\hat{f}_h(8.5) = 0.067$
The uniform kernel jumps because a point is either counted or not counted, the Gaussian kernel changes smoothly because the weight of a point slowly decreases as it gets further away

d) I would prefer the Gaussian kernel because it gives a smoother estimate, because the weight gradually decrease avoids sudden changes when x moves slightly

Problem #4

problems/q4.ipynb